Sprint 4 — Backend and Technical Decisions

1. Sprint 4 Incremental Purpose

This document records only the new AI-related design, backend architecture decisions and software implementation decisions introduced for Sprint 4. The Sprint 2 and Sprint 3 requirements remain frozen and are not restated or redesigned here. The backend design shall preserve the Sprint 2 and Sprint 3 requirements.

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

This document covers Gemini, Flutter, Dart, Visual Studio Code, the supporting Windows role of Visual Studio, and the backend and implementation technologies described in the following sections.

---

2. AI Integration Design

OUTDATED — Superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. Retained for historical traceability. Current rule: Qwen3.5-9B is the primary model, Qwen3.5-4B is only the manually configured development and low-spec alternative, Ollama is the current runtime, and Qwen Local Model Adapter replaces the Gemini Provider Adapter.

2.1 Gemini Decision

OUTDATED — Superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. Retained for historical traceability. Current rule: Qwen3.5-9B is the primary model, Qwen3.5-4B is only the manually configured development and low-spec alternative, Ollama is the current runtime, and Qwen Local Model Adapter replaces the Gemini Provider Adapter.

Gemini is the AI provider for the project.

OUTDATED — Superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. Retained for historical traceability. Current rule: Qwen3.5-9B is the primary model, Qwen3.5-4B is only the manually configured development and low-spec alternative, Ollama is the current runtime, and Qwen Local Model Adapter replaces the Gemini Provider Adapter.

The team intends to use a Gemini access option that provides a free usage allowance available to the team. This document does not assume that the allowance is permanent, unlimited or suitable for future production use. The exact Gemini model name and version have not yet been specified and shall remain configurable.

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

Gemini is used only for approved AI-supported functions. Its role is limited to producing constrained language-based output from data already selected and permitted by the backend.

2.2 Responsibility Boundary

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

The Flutter client shall not call Gemini directly. All Gemini requests shall pass through the backend.

The backend is responsible for:

- authenticating the user;
- identifying the approved AI request purpose;
- checking user permissions and permitted data use;
- retrieving only the information required for the current request;
- minimising and structuring the AI context;

OUTDATED — Superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. Retained for historical traceability. Current rule: Qwen3.5-9B is the primary model, Qwen3.5-4B is only the manually configured development and low-spec alternative, Ollama is the current runtime, and Qwen Local Model Adapter replaces the Gemini Provider Adapter.

- calling Gemini through a server-side adapter;

OUTDATED — Superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. Retained for historical traceability. Current rule: Qwen3.5-9B is the primary model, Qwen3.5-4B is only the manually configured development and low-spec alternative, Ollama is the current runtime, and Qwen Local Model Adapter replaces the Gemini Provider Adapter.

- validating Gemini output;
- applying deterministic fallback behaviour;
- returning only an approved result to the Flutter client.

OUTDATED — Superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. Retained for historical traceability. Current rule: Qwen3.5-9B is the primary model, Qwen3.5-4B is only the manually configured development and low-spec alternative, Ollama is the current runtime, and Qwen Local Model Adapter replaces the Gemini Provider Adapter.

Gemini may support:

- natural-language recommendation explanations;
- summarisation of permitted context;
- constrained suggestions;
- approved AI activity recommendations;
- approved AI Check-in assistance.

OUTDATED — Superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. Retained for historical traceability. Current rule: Qwen3.5-9B is the primary model, Qwen3.5-4B is only the manually configured development and low-spec alternative, Ollama is the current runtime, and Qwen Local Model Adapter replaces the Gemini Provider Adapter.

Gemini shall not be responsible for:

- authentication;
- direct database queries;
- final candidate selection;
- permission or visibility decisions;
- direct database writes;
- friendship establishment;
- modification of Profile visibility;
- creation of users or events;
- critical application-state changes.

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

Gemini shall not directly access the application database. Under the selected backend architecture, Cloud Firestore shall be accessed only through the backend service and data-access layers. Gemini receives only the structured and minimised context prepared for the current request and shall not directly query or write application data.

2.3 Simplified AI Request Flow

1. The Flutter client sends an authenticated AI request to the backend.
2. The backend identifies the approved request purpose.
3. The backend loads only permitted and required data.
4. The backend minimises and structures the context.

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

5. The backend calls Gemini through a server-side adapter.

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

6. The backend validates the Gemini response.
7. The backend returns only the validated result.

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

8. A deterministic fallback is used if Gemini is unavailable or invalid.

2.4 Response Validation

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

The backend shall validate Gemini output before it is returned to the client or used by another application component.

Validation should check:

- the expected response format;
- required fields;
- allowed identifiers;
- allowed recommendation factors;
- missing or malformed values;
- restricted information;
- unsupported claims;
- output that falls outside the approved request purpose.

OUTDATED — Superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. Retained for historical traceability. Current rule: Qwen3.5-9B is the primary model, Qwen3.5-4B is only the manually configured development and low-spec alternative, Ollama is the current runtime, and Qwen Local Model Adapter replaces the Gemini Provider Adapter.

Where candidate identifiers are supplied to Gemini, the model may refer only to those approved candidates. Unknown users, events or recommendation factors shall be rejected rather than accepted as new application data.

The exact structured response format remains an implementation decision. However, the backend should use a predictable machine-readable response rather than allowing unrestricted free-form output to directly control application behaviour.

2.5 Failure and Fallback

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

A deterministic fallback shall be available when Gemini cannot provide a valid result.

Fallback may be used for:

- timeout;
- network failure;
- rate limiting;
- invalid response format;
- missing required fields;
- unsupported output;

OUTDATED — Superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. Retained for historical traceability. Current rule: Qwen3.5-9B is the primary model, Qwen3.5-4B is only the manually configured development and low-spec alternative, Ollama is the current runtime, and Qwen Local Model Adapter replaces the Gemini Provider Adapter.

- unavailable Gemini service.

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

The fallback shall use only approved application data and shall not invent users, events, permissions or application states. Core non-AI functions shall remain usable when Gemini is unavailable.

The exact timeout, retry and rate-limit values will be configured during implementation and testing. Any retry behaviour should remain controlled and should not cause repeated provider requests without a defined limit.

2.6 AI Data and Logging Boundary

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

AI context shall contain only data needed for the current approved request purpose. The backend shall apply permission checks and data minimisation before sending the context to Gemini.

OUTDATED — Superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. Retained for historical traceability. Current rule: Qwen3.5-9B is the primary model, Qwen3.5-4B is only the manually configured development and low-spec alternative, Ollama is the current runtime, and Qwen Local Model Adapter replaces the Gemini Provider Adapter.

Raw prompts and raw Gemini responses should not be stored for long-term use by default. Audit records should primarily contain non-sensitive operational metadata, such as:

- request purpose;
- request identifier;
- response category;
- validation result;
- fallback usage;
- latency;
- configured model reference.

OUTDATED — Superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. Retained for historical traceability. Current rule: Qwen3.5-9B is the primary model, Qwen3.5-4B is only the manually configured development and low-spec alternative, Ollama is the current runtime, and Qwen Local Model Adapter replaces the Gemini Provider Adapter.

Gemini credentials shall not be stored in the Flutter client or committed to the source repository. The retention period for AI-related operational records will be configured during implementation.

---

3. Backend Architecture

The backend platform is Firebase.

Firebase provides a managed serverless backend suitable for the current Flutter student prototype. It reduces the need to configure and maintain a self-hosted server while providing services that can support authentication, persistent application data, protected server-side execution and local development tools.

This design does not include physical servers, Kubernetes, microservices or complex enterprise infrastructure.

3.1 Server-Side Execution

Firebase Cloud Functions are used for protected server-side operations.

Cloud Functions:

- receive authenticated backend requests;
- validate request data;
- execute permission-sensitive operations;

OUTDATED — Superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. Retained for historical traceability. Current rule: Qwen3.5-9B is the primary model, Qwen3.5-4B is only the manually configured development and low-spec alternative, Ollama is the current runtime, and Qwen Local Model Adapter replaces the Gemini Provider Adapter.

- prepare Gemini context;

OUTDATED — Superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. Retained for historical traceability. Current rule: Qwen3.5-9B is the primary model, Qwen3.5-4B is only the manually configured development and low-spec alternative, Ollama is the current runtime, and Qwen Local Model Adapter replaces the Gemini Provider Adapter.

- call Gemini;
- validate AI responses;
- return controlled results to the Flutter client.

TypeScript on a supported Node.js runtime is used as the backend function language. Flutter and Dart remain the client technologies; selecting TypeScript for Cloud Functions does not mean that the whole project uses TypeScript.

3.2 Database and Authentication

Cloud Firestore is the operational database.

Its responsibilities are limited to:

- storing application runtime data;
- supporting persistence for backend services;
- integrating with Firebase and Flutter;
- operating with backend permission checks and Firebase Security Rules.

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

Cloud Firestore access for protected application operations shall pass through the backend service and data-access layers. Gemini shall not directly query or write Cloud Firestore.

This Sprint 4 document does not define a complete collection structure, field catalogue or ERD.

Firebase Authentication with email and password authentication is used for authentication.

Social login, SingPass implementation, biometric login and anonymous login are not introduced.

3.3 API Approach

Protected backend operations should use HTTPS Cloud Functions or Firebase Callable Functions with authenticated JSON requests and responses.

The API layer should:

- verify authentication;
- validate request input;
- call the appropriate service;
- return consistent success or error responses;
- avoid exposing backend secrets or internal implementation details.

A complete endpoint catalogue is intentionally excluded from this incremental document.

3.4 Serverless Layered Architecture

The backend uses a simple serverless layered architecture:

1. **API / Function Layer**
   Receives authenticated requests, validates inputs and calls the appropriate service.

2. **Service Layer**
   Applies business rules, performs permission checks, prepares permitted AI context and processes validated AI results.
3. **Data Access Layer**
   Isolates Cloud Firestore reads and writes from the API and service layers.

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

4. **External AI Integration Layer**
   Contains the Gemini Provider Adapter, server-side Gemini request logic, response validation and fallback handling.

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

This structure keeps the backend understandable for a student project while preventing controllers, database access and Gemini integration from being mixed into one function.

3.5 Secret Management

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

Gemini credentials and other sensitive configuration should be stored in backend environment configuration or a supported secret-management service.

Production secrets shall not be:

- embedded in the Flutter client;
- committed to GitHub;
- written into normal application logs;
- returned through backend responses.

---

4. Software Implementation Decisions

Flutter and Dart are the client framework and language.

Visual Studio Code with the Flutter and Dart extensions is the primary environment for code authoring and Flutter debugging.

Visual Studio is a supporting Windows tool only. It may be required for:

- Windows build toolchains;

- Windows desktop support;
- Windows-specific debugging;
- native dependencies required by Flutter on Windows.

It is not the primary Flutter IDE.

4.1 Client Architecture

The Flutter client uses a layered client architecture with a Repository pattern.

The structure contains:

1. **Presentation Layer**
   - screens;
   - widgets;
   - UI state;
   - user interaction.
2. **Application / Service Layer**
   - application logic;
   - validation;
   - coordination between the UI and repositories;
   - backend-service calls.
3. **Data Layer**
   - repositories;
   - Firebase access;
   - backend API clients;
   - data mapping.

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

This structure separates UI code from application logic and data access, improves maintainability and supports testing. Repositories isolate data access from the other client layers. UI widgets should not directly call Gemini, manage backend secrets or contain protected backend logic.

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

The client architecture remains limited to a simple structure suitable for the current student prototype. Final permission decisions, Gemini access and other protected operations remain backend responsibilities.

4.2 Source Control

Git and GitHub are used for source control and team collaboration.

The workflow uses:

- feature branches;
- focused commits;
- pull requests;
- review before merging;
- exclusion of secrets and local environment files from the repository.

4.3 Testing Support

The testing support includes:

- Flutter unit tests;
- Flutter widget tests;
- backend function tests;
- Firebase Emulator Suite;
- manual prototype verification.

The Firebase Emulator Suite supports local testing of Firebase services without relying entirely on hosted data. This document does not define UAT scenarios or a complete test plan.

---

5. Technical Decision Summary

OUTDATED — Superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. Retained for historical traceability. Current rule: Qwen3.5-9B is the primary model, Qwen3.5-4B is only the manually configured development and low-spec alternative, Ollama is the current runtime, and Qwen Local Model Adapter replaces the Gemini Provider Adapter.

Decision Area	Technology / Approach	Purpose
AI Provider	Gemini	Supports approved AI explanations, summaries and constrained suggestions.
Client Framework	Flutter	Provides the cross-platform application client.
Client Language	Dart	Implements the Flutter client.
Primary IDE	Visual Studio Code	Primary code-authoring and Flutter debugging environment.

Supporting Windows Tool	Visual Studio	Supports Windows toolchains, desktop builds and native dependencies.
Backend Platform	Firebase	Provides a managed serverless backend for the prototype.

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

Server-side Functions	Firebase Cloud Functions	Runs protected backend and Gemini integration operations.
Backend Function Language	TypeScript / Node.js	Implements Cloud Functions.
Database	Cloud Firestore	Stores persistent application runtime data under backend service and data-access control.
Authentication	Firebase Authentication	Provides email and password identity management.
API Approach	HTTPS or Callable Cloud Functions	Supports authenticated JSON backend requests.
Client Architecture	Layered client architecture with Repository pattern	Separates UI, application logic and data access.
Backend Architecture	Serverless layered architecture	Separates functions, services, data access and AI integration.

PARTIALLY OUTDATED — Only the Gemini-specific provider, model, API, credential, SDK, quota, billing, and Adapter wording is superseded by Sprint 5 NIS-CH01, NIS-CH02, NIS-AC01–NIS-AC07, and NIS-FR01–NIS-FR10. The remaining Backend-only architecture, validation, privacy, data minimisation, no-direct-database-access, and deterministic Backend fallback content is still applicable.

Secret Management	Backend environment secrets	Protects Gemini credentials and sensitive configuration.
Source Control	Git and GitHub	Supports collaboration, branching and review.

Testing Support	Flutter tests and Firebase Emulator Suite	Supports client, backend and local integration testing.